fn accuracy_to_discrete_gaussian_scale

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This document contains materials associated with accuracy_to_discrete_gaussian_scale.
By discrete_gaussian_scale_to_accuracy, the relationship between α, a and $scale$, is:

$$1 - \alpha = \frac{\sum_{y=0}^{a-1}(1 + 1[y \neq 0])e^{-(y/s)^2/2}}{\sum_{z \in \mathbb{Z}} e^{-(z/s)^2/2}}$$

A closed-form expression for s doesn't exist, so we use a numerical approach by a binary search. A loose upper bound is provided by accuracy_to_gaussian_scale. The binary search finds the smallest s such that

$$\alpha \leq 1 - \frac{\sum_{y=0}^{a-1}(1 + 1[y \neq 0])e^{-(y/s)^2/2}}{\sum_{z \in \mathbb{Z}} e^{-(z/s)^2/2}}$$